

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283710

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Cyr; Gregory et al.

TAPE MEASURE WITH POP OUT BLADE PRESENTATION

Abstract

A retractable rule assembly comprises a housing, a reel, an elongated blade, a hook, a coil spring, and a blade advancer. The reel is rotatably mounted in the housing. The elongated blade is wound on the reel and is constructed and arranged to extend outwardly through an opening in the housing. The hook is positioned on a distal end of the blade. The coil spring is configured to rotate the reel in the housing assembly in a direction to wind up the elongated blade when extending outwardly of the housing. The blade advancer is constructed and arranged to be manually engaged and moved relative to the housing and into contact with the blade and/or the reel. The blade advancer is movable against a bias of the coil spring to advance the blade in a direction that extends the blade outwardly from the housing.

Inventors: Cyr; Gregory (Oakville, CT), Schafer; Joshua P. (Wolcott, CT), Jaiswal; Somen Rajkumar (Danbury, CT), Novak; Joseph T. (East Longmeadow, MA), Green; Matthew Christopher (Amherst, MA), Green; Benjamin J. (Stamford, CT), Lu; Ke (San Diego, CA)

Applicant: Stanley Black & Decker, Inc. (New Britain, CT)

Family ID: 94532921

Appl. No.: 18/601392

Filed: March 11, 2024

Publication Classification

Int. Cl.: G01B3/1007 (20200101); G01B3/1005 (20200101); G01B3/102 (20200101); G01B3/1056 (20200101)

U.S. Cl.:

CPC G01B3/1007 (20200101); G01B3/102 (20200101); G01B3/1056 (20130101); G01B2003/103 (20130101); G01B2003/1038 (20130101)

Background/Summary

FIELD

[0001] The present patent application relates to tape measures.

BACKGROUND

[0002] A tape measure or tape rule assembly is a measuring device. The tape measure generally includes an elongated rule blade that is mounted on a reel rotatably disposed within a housing. The blade may be a flexible tape or a retractable tape. The blade may be resiliently coiled within the housing. That is, the blade may be spring biased into the housing. The blade is normally printed with incremental measuring marks/indicia for measuring distances. The blade is retracted into the housing for storage by coiling the blade about the reel. A front of the blade generally includes a hook or an end hook that provides a surface to engage and fix the end of the blade against a point to be measured from, so the measurement can be made from the hook to a position on the blade indicated by the measuring indicia. The housing may often have a lock mechanism that allows the blade to be locked into a certain position relative to the housing (e.g., locked into an extended position, preventing the spring bias from retracting the blade into the housing). The housing may also often have a belt clip located thereon that allows users to clip the tape measure onto their belt. Further, in the tape measure assembly, the blade is typically fixed to a motor assembly that includes the spring therein and is tensioned against a fixed hub. Traditional springs are attached to reels or cases in tape measure via physical interlocks, cut into one or more of the spring and an attaching surface, or are assembled with physical retainers such as rivets, screws, or bolts.

[0003] To measure a workpiece, a length of the blade is pulled out of the housing to span the distance to be measured and the blade or hook is held against the workpiece so that gradation lines and numbers printed on the blade can be read against a point on the workpiece. To measure a distance between two objects or surfaces, the hook at the free end of the blade may be temporarily secured or placed against an object or surface. Distances can then be marked with a marking tool, such as a pencil, directly onto the surface measured.

[0004] The present patent application provides improvements in the tape measures or retractable rule assemblies.

SUMMARY

[0005] The present patent application provides a retractable rule assembly.

[0006] One aspect of the present patent application provides a retractable rule assembly. The retractable rule assembly comprises a housing, a reel, an elongated blade, a hook, a coil spring, and a blade advancer. The reel is rotatably mounted in the housing. The elongated blade is wound on the reel and is constructed and arranged to extend outwardly through an opening in the housing. The hook is positioned on a distal end of the blade. The coil spring is configured to rotate the reel in the housing assembly in a direction to wind up the elongated blade when extending outwardly of the housing. The blade advancer is constructed and arranged to be manually engaged and moved relative to the housing and into contact with the blade and/or the reel. The blade advancer is movable against a bias of the coil spring to advance the blade in a direction that extends the blade

outwardly from the housing.

[0007] Implementations of the foregoing aspects may include one or more of the following features.

[0008] In an aspect, the blade advancer may comprise a manually engageable roller.

[0009] In an aspect, the retractable rule assembly may further comprise a spring that biases the roller away from the blade and/or reel. The roller may be configured to be manually forced against the bias of the spring and into engagement with the blade and/or reel and manually rolled to move the blade in the direction that extends the blade outwardly from the housing.

[0010] In an aspect, the roller may be disposed above the blade and is normally held away from the blade due to the bias of the spring

[0011] In an aspect, the roller may be disposed below the blade and is normally held away from the blade under the force of gravity.

[0012] In an aspect, the blade advancer may form part of a releasable blade lock assembly. The blade lock assembly may be constructed and arranged to be manually actuated to lock the blade in any position of extension outwardly of the housing opening and to release the blade from any position in which it is locked.

[0013] In an aspect, the blade lock assembly may be movable in a first direction to lock the blade and a second direction, opposite direction, to advance the blade.

[0014] In an aspect, the blade advancer may include a first engaging portion disposed on the blade lock assembly and a second engaging portion disposed on the reel. The first engaging portion of the blade lock assembly may be configured to interlock with the second engaging portion of the reel to facilitate advancement of the blade in the direction that extends the blade outwardly from the housing. The first engaging portion may include a projection and the second engaging portion may include teeth that are configured to engage and interlock with the projection.

[0015] In an aspect, the blade advancer, which forms part of releasable blade lock assembly, may include a cam surface. The cam surface may be configured to force a friction pad feature against the blade/reel to facilitate advancement of the blade in the direction that extends the blade outwardly from the housing.

[0016] In an aspect, the blade advancer, which forms part of releasable blade lock assembly, may include an end hook pusher. The end hook pusher may be configured to engage with the hook positioned on the distal end of the blade to push the hook and the blade connected thereto outwardly from the housing.

[0017] In an aspect, the blade advancer may include a manually engageable portion disposed on the housing and moveable along a slot disposed on the housing, a drive gear operatively connected to the manually engageable portion, and a driven gear operatively connected to the reel. The manually engageable portion may include a rest configuration and an operational configuration. When the manually engageable portion is in its rest configuration, there may be no engagement between the drive gear and the driver gear. When the manually engageable portion is in its operational configuration, the drive gear and the driver gear may be configured to engage with each other so as to rotate the reel.

[0018] In an aspect, when a force is applied on the manually engageable portion, the manually engageable portion may be moved along the slot from its rest configuration to its operational position. When the manually engageable portion is in its operational configuration, the drive gear and the driver gear may be configured to engage with each other to facilitate the rotation of the reel and the advancement of the blade in the direction that extends the blade outwardly from the housing.

[0019] In an aspect, the manually engageable portion may include a spring-biased manually engageable lever.

[0020] Another aspect of the present patent application provides a retractable rule assembly comprises a housing, a reel, an elongated blade, a coil spring, and a spring biased blade advancer.

The reel is rotatably mounted in the housing. The elongated blade is wound on the reel and is constructed and arranged to extend outwardly through an opening in the housing. The coil spring is configured to rotate the reel in the housing assembly in a direction to wind up the elongated blade when extending outwardly of said housing. The spring biased blade advancer comprises a pusher and a spring, the blade advancer being constructed and arranged such that when actuated, the spring pushes the pusher to engage a portion of the blade or the reel against a bias of the coil spring to advance the blade in a direction that extends the blade outwardly from the housing.

[0021] Implementations of the foregoing aspects may include one or more of the following features.

[0022] In an aspect, the blade may comprise a hook. The pusher, when moved by the spring, may contact the hook to extend the blade outwardly from the housing.

[0023] In an aspect, a spring force of the blade advancer spring may be sufficiently large to overcome the spring force of the coil spring retracting the blade into the housing.

[0024] In an aspect, the blade advancer may remain extended until manually forced to a retracted position.

[0025] In an aspect, the blade advancer may serve as a shock absorber against rapid retraction of the blade while in the extended position.

[0026] In an aspect, the blade advancer may serve as additional finger support while in the extended position.

[0027] These and other aspects of the present patent application, as well as the methods of operation and functions of the related elements of structure and the combination of parts and economies of manufacture, will become more apparent upon consideration of the following description and the appended claims with reference to the accompanying drawings, all of which form a part of this specification, wherein like reference numerals designate corresponding parts in the various figures. In one embodiment of the present patent application, the structural components illustrated herein are drawn to scale. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only and are not intended as a definition of the limits of the present patent application. It shall also be appreciated that the features of one embodiment disclosed herein can be used in other embodiments disclosed herein. As used in the specification and in the claims, the singular form of “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise. In addition, as used in the specification and the claims, the term “or” means “and/or” unless the context clearly dictates otherwise. It should also be appreciated that some of the components and features discussed herein may be discussed in connection with only one (singular) of such components, and that additional like components which may be disclosed herein may not be discussed in detail for the sake of reducing redundancy.

[0028] Other aspects, features, and advantages of the present patent application will become apparent from the following detailed description, the accompanying drawings, and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] Various embodiments are disclosed, by way of example only, with reference to the accompanying schematic drawings in which corresponding reference symbols indicate corresponding parts, in which

[0030] FIGS. 1A and 1B show a partial exploded perspective view and a fully exploded view of a retractable rule assembly according to an embodiment of the present patent application, wherein a top manually engageable roller acting as a blade advancer is shown;

[0031] FIG. 1C shows a cross-sectional view of the retractable rule assembly, wherein only one

housing member and a releasable blade lock assembly are being shown (and other components of the retractable rule assembly are not shown for sake of clarity and to better illustrate the housing member and the releasable blade lock assembly);

[0032] FIG. 2 shows a partial (top portion) perspective view of a housing portion of the retractable rule assembly according to an embodiment of the present patent application, wherein the top manually engageable roller acting as the blade advancer and a spring that biases the roller away from a blade and/or a reel are disposed in the housing portion;

[0033] FIGS. 3 and 4 show a partial (top portion) perspective view and a partial top view of the housing portion of the retractable rule assembly according to an embodiment of the present patent application, wherein the top manually engageable roller acting as the blade advancer and the spring that biases the roller away from the blade and/or a reel are not shown so that a spring receiving opening configured to receive the spring and a roller receiving portion configured to receive a portion of the roller are clearly shown;

[0034] FIG. 5 shows a partial perspective view of the retractable rule assembly according to an embodiment of the present patent application, wherein a bottom manually engageable roller acting as the blade advancer is shown;

[0035] FIG. 6 shows a partial (bottom portion) perspective view of the housing portion of the retractable rule assembly according to an embodiment of the present patent application, wherein the bottom manually engageable roller acting as the blade advancer is shown as being disposed in the housing portion;

[0036] FIG. 7 shows another partial (bottom portion) perspective view of the housing portion of the retractable rule assembly according to an embodiment of the present patent application, wherein the bottom manually engageable roller acting as the blade advancer is not shown so that a roller supporting portion configured to support a portion of the roller thereon is clearly shown, wherein portions of the reel mounted in the housing portion are also shown in FIG. 7;

[0037] FIG. 8 shows a partial (bottom portion) cross-sectional view of the retractable rule assembly according to an embodiment of the present patent application, wherein the bottom manually engageable roller acting as the blade advancer is shown as being disposed in the housing portion;

[0038] FIG. 9 shows an exemplary view of the retractable rule assembly according to an embodiment of the present patent application, wherein the bottom manually engageable roller acting as the blade advancer and the reel are shown adjacent to each other;

[0039] FIGS. 10A-10D show various views of the retractable rule assembly according to an embodiment of the present patent application, wherein the blade advancer may form part of releasable blade lock assembly, wherein the releasable blade lock assembly is shown in a locked configuration in FIG. 10A and an unlocked configuration in FIG. 10B, and wherein the blade advancer is shown in a spool engaged configuration in FIG. 10C and in a blade extended configuration in FIG. 10D in which the blade is advanced in a direction that extends the blade outwardly from the housing;

[0040] FIGS. 11A-11D show various views of the retractable rule assembly according to another embodiment of the present patent application, wherein the blade advancer may form part of releasable blade lock assembly, wherein the releasable blade lock assembly is shown in a locked configuration in FIG. 11A and an unlocked configuration in FIG. 11B, and wherein the blade advancer is shown in a friction pad engaged configuration in FIG. 11C and in a blade extended configuration in FIG. 11D in which the blade is advanced in the direction that extends the blade outwardly from the housing;

[0041] FIGS. 12A-12D show various views of the retractable rule assembly according to yet another embodiment of the present patent application, wherein the blade advancer may form part of releasable blade lock assembly, wherein the releasable blade lock assembly is shown in a locked configuration in FIG. 12A and an unlocked configuration in FIG. 12B, and wherein the blade advancer is shown in an extension lock disengaged or released (prior to being pressed) and engaged

(when pressed) configuration in FIG. 12C and in a blade extended configuration in FIG. 12D in which the blade is advanced in the direction that extends the blade outwardly from the housing; [0042] FIG. 13 shows a perspective view of a housing portion of the retractable rule assembly according to an embodiment of the present patent application, wherein the blade advancer may have a micro-SD card spring action configuration to extend the blade, wherein a blade feeder and an extension spring of the blade advancer are shown in FIG. 13; [0043] FIGS. 14 and 15 show the micro-SD card spring action configuration that may be used by the blade advancer, according to an embodiment of the present patent application, to advance the blade in the direction that extends the blade outwardly from the housing; and [0044] FIGS. 16 and 17 show a retractable rule assembly that may be configured to automatically extend the blade with a press of a button or the engagement of a lever according to an embodiment of the present patent application.

DETAILED DESCRIPTION

[0045] The present patent application provides a retractable rule assembly 10 as shown in FIGS. 1A-1C. The rule assembly 10 includes a housing 12, a reel 14, an elongated blade 16, a hook (or end hook) 20, a coil spring 22, and a blade advancer 24. The reel 14 may be rotatably mounted in the housing 12. The elongated blade 16 may be wound on the reel 14 and may be constructed and arranged to extend outwardly through an opening 18 in the housing 12. The hook 20 may be positioned on a distal end 28 of the blade 16. The coil spring 22 may be configured to rotate the reel 14 in the housing 12 in a direction to wind up the elongated blade 16 when extending outwardly of the housing 12. The blade advancer 24 may be constructed and arranged to be manually engaged and moved relative to the housing 12 and into contact with the blade 16 and/or the reel 14. The blade advancer 24 may be movable against a bias of the coil spring 22 to advance the blade 16 in a direction that extends the blade 16 outwardly from the housing 12.

[0046] The retractable rule assembly 10 may interchangeably referred to as tape measure or tape rule case/assembly. The present patent application provides the blade advancer 24 for presenting the tape blade 16 out of (and away from) the housing 12 so a user can more easily grab the end hook 20 and start extending the blade 16 away from the housing 12. For example, the blade advancer 24 of the present patent application may include or be in the form of a blade roller (including a top manually engageable roller as shown in and described in detail below with respect to FIGS. 1A-4 and/or a bottom manually engageable roller as shown in and described in detail below with respect to FIGS. 5-9), a dual action slidelock (having three different configurations as shown in and described in detail below with respect to FIGS. 10A-12D), or may have a microSD (i.e., a type of removable flash memory card used for storing information) type spring action configuration (as shown in and described in detail below with respect to FIGS. 13-15). The blade advancer 24 of the present patent application may have the automatic extension configuration as shown in and described with respect to FIGS. 16-17.

[0047] The housing 12 may be constructed to easily and comfortably fit in a hand of the user because it optimizes the use of space within the housing 12 to house the blade 16, the reel 14, the coil spring 22 including a spindle/shaft 32, and other components of the retractable rule assembly 10. The housing 12 may be constructed of, for example, a molded plastic material.

[0048] The housing 12 may include a pair of cooperating housing members 12a, 12b (e.g., a first housing member 12a and a second housing member 12b). The housing members 12a, 12b may include portions along abutting free edges thereof. The portions may help secure the molded housing members 12a, 12b of the assembled rule assembly 10 together. The second housing member 12b and the first housing member 12a may be movable toward one another in an axial direction into cooperating relation to define the housing 12. The "axial direction" may refer to the direction of the axis of rotation of the reel 14 defined by the shaft 32.

[0049] Each housing member 12a, 12b may include four corner portions 36, 38, 40, 42. For example, the first corner portion 36 may be adjacent the blade opening 18, the second corner

portion **38** may be at an opposite bottom end of the housing **12**, and the third and fourth corner portions **40** and **42** may be in peripheral top portions of the housing **12**. Each corner portion **36**, **38**, **40**, **42** of each housing member **12a**, **12b** may include either a recess or a projection and an opening in the respective recess/projection. At each corner portion **36**, **38**, **40**, **42**, the recess/projection of one of the housing members **12a**, **12b** may be configured to receive or engage with the projection/recess of the other of the housing member **12a**, **12b**. Also, the opening in the recess/projection of one of the housing members **12a**, **12b** may be configured to align with the opening in the projection/recess of the other of the housing members **12a**, **12b**. Fasteners **44**, **46**, **48**, **50** may then be positioned in aligned openings of the corner portions **36**, **38**, **40**, **42** of the housing member **12a**, **12b**. Thus, it can be appreciated that the housing members **12a**, **12b** may be secured together using fasteners in these locations.

[0050] A belt clip **30** may be secured to one side of the housing **12** by a fastener **138** (as shown in FIG. 5). The belt clip **30** may be generally used to attach the rule assembly **10** to the belt of a user, or other attachment point. The belt clip **30** may be made from, for example, a metal material.

[0051] The reel **14** may be interchangeably referred to as spool or blade reel. The reel **14** may have a first end portion **52** and an opposing second end portion **54**. The reel **14** may include a cylindrical/peripheral side wall **56** extending between the first end portion **52** and the second end portion **54**. The second end portion **54** of the reel **14** may include a second wall **58** extending perpendicular to the cylindrical side wall **56**. The second wall **58** may form base/bottom wall of the reel **14**. The second wall **58** may also extend perpendicular to and outwardly away from the cylindrical side wall **56** to form a flange portion **60**. The second wall **58** and the cylindrical side wall **56** of the reel **14** may together define a storage space **SS** and an access opening **AO** to the storage space **SS**. A first wall **62** may be detachably/removably connected to the circumferential side wall **56** at the first end **52**. The first wall **62** may be configured to cover the storage space **SS** and the access opening **AO**. The first wall **62** may form top wall/lid/cover of the reel **14**. The first wall **62** may also extend perpendicular to and outwardly away from the cylindrical side wall **56** to form a flange portion **64**. As will be clear from the discussions below, the coil spring **22** and at least a portion of the shaft **32** are received in the storage space **SS**. The reel **14**, including the first wall **62**, the second wall **58** and the cylindrical side wall **56**, may have a cylindrical shaped configuration.

[0052] The first wall **62** may include an opening **66**. The opening **66** of the first wall **62** may be disposed centrally (i.e., in a central portion **CP**) in the first wall **62**. The opening **66** of the first wall **62** may be configured to allow a first end portion **68** of the shaft **32** to pass therethrough such that the first end portion **68** of the shaft **32** may be fixedly connected to a portion (not shown but may be disposed centrally (i.e., in a central portion) on the housing member **12a**) of the housing **12**.

[0053] Similarly, the second wall **58** may include an opening (not shown). The opening of the second wall **58** may be disposed centrally (i.e., in a central portion) in the second wall **58**. The opening of the second wall **58** may be configured to allow a second end portion **70** of the shaft **32** to pass therethrough such that the second end portion **70** of the shaft **32** may be fixedly connected to a portion **72** (disposed centrally (i.e., in a central portion) on the housing member **12b**) of the housing **12**.

[0054] The reel **14** may be made of a molded plastic and may be provided with a slot or an opening **74** in the cylindrical side wall **56** thereof. A first longitudinal end (not shown but may be opposite to the second longitudinal free end/distal end **28**) of the blade **16** may terminate in a hook-like structure that engages a second longitudinal end of the coil spring **22** (disposed in the storage space **SS** of the reel **14**) so as to connect the first longitudinal end of the blade **16** to the second longitudinal end of the coil spring **22**. The coil spring **22** may have the second opposing longitudinal end and an opposing first longitudinal end. The second longitudinal free end (or the distal end) **28** of the blade **16** may extend generally outwardly of the reel **14**.

[0055] The reel **14** may have an outer wall containing the blade **16**. The reel **14** may be configured

to enclose the blade **16** on three sides. For example, the reel **14** may have outer surfaces (e.g., of the cylindrical side wall **56**, of the flange portion **64** of the first wall portion **62**, and of the flange portion **60** of the second wall portion **58**) that are configured to contain the blade **16**.

[0056] The elongated blade or blade **16** may be formed of a ribbon of metal (e.g., the metal being steel), and a top concave surface of the blade **16** may be printed with measuring lines and digits (not shown) for measuring lengths and distances. The blade **16** may include the first longitudinal end (that may be secured to the coil spring **22** via the slot/opening **74**) and the opposing, second longitudinal free end **28** of the blade **16** may extend generally outwardly of the housing **12**. The blade **16** may be constructed and arranged with respect to the housing **12** to extend generally from a position tangential of the reel **14** outwardly through the blade opening **18** provided in the housing **12**.

[0057] The blade **16** may be generally movable between a fully retracted position to a fully extended position. It can be appreciated that as the blade **16** is unwound from the reel **14**, the coil spring **22** may be wound around the rigidly fixed shaft **32**. This winding of the coil spring **22** around the shaft **32** stores energy in the coil spring **22** to provide spring powered rewinding of the blade **16** around the reel **14** when the extended blade **16** is released.

[0058] The blade **16** may be constructed of a ribbon of sheet metal that is shaped during the manufacturing to have a normal or memory configuration that has a generally arcuate or concavo-convex transverse cross-section. When a portion of the blade **16** is wound about the reel **14**, the wound portion has a flat transverse cross-section and the wound layers of the coiled blade provide the wound blade with an abutting volute coil configuration. In other words, when the blade **16** is wound around the reel **14**, it has the flat cross-section and when the blade **16** is withdrawn from the housing **12** to measure an object, it returns to the concavo-convex cross-section. Thus, the coil spring **22** is constructed and arranged to rotate the reel **14** about the shaft **32** with respect to the housing **12** in a direction to wind up the elongated blade **16** when extending outwardly of the blade opening **18** in a normal concavo-convex cross-sectional configuration onto the reel **14** in an abutting volute coil formation in a flattened cross-sectional configuration. The concavo-convex cross-section provides the extended blade with rigidity and maintains the blade essentially straight in the longitudinal direction. The concavo-convex cross-section of the blade **16** generally provides the unsupported blade **16** with blade standout. The concavo-convex cross-section of the blade **16** generally includes an arcuate central section and integral end sections. In various embodiments, the concavo-convex cross-sectional configuration of the blade **16** can be of the type described in commonly assigned U.S. Pat. No. 6,324,769, which is hereby incorporated by reference in its entirety.

[0059] The blade **16** may have the end hook **20** on the free/distal end **28** thereof. The end hook **20** may include a mounting portion **76** and a hook portion **78** bent at a generally right angle from an end of the mounting portion **76**. In one embodiment, the end hook **20** is formed of, for example, a sheet metal material of a predetermined thickness. The hook portion **78** may have a generally U-shaped configuration. The mounting portion **76** may have a generally concavo-convex configuration. The mounting portion **76** may have a generally flat configuration. The end hook **20** may be mounted on the free end **28** of the blade **16** with the mounting portion **76** thereof secured in engagement with a concave (upper) side of the free end **28** of the blade **16** and in overlying relation thereto. In one embodiment, the connection between the free end **28** of the blade **16** and the mounting portion **76** may permit limiting sliding movement therebetween. For example, such a configuration of the limited sliding movement of the end hook **20** with respect to the blade **16** can be of the type described in commonly assigned U.S. Pat. No. 8,528,222, which is hereby incorporated by reference in its entirety.

[0060] The coil spring **22** may be constructed and arranged to be received and stored within (the storage space **SS** of) the reel **14**. The coil spring **22** may be constructed and arranged to rotate the reel **14** with respect to the housing **12** in a direction to wind the elongated blade **16** about the reel

14 when the blade **16** is extending outwardly of the blade opening **18**. The coil spring **22** may include the first longitudinal end and the opposing second longitudinal end. The first longitudinal end of the coil spring **22** may be secured to the shaft **32** via a slot (not shown) in the shaft **32**, and the second longitudinal end of the coil spring **22** may be secured to the blade **16** via slot **74** in the reel **14**. The coil spring **22** may be a thin, flat ribbon of metal (e.g., the metal being steel).

[0061] The shaft **32** may be constructed of, for example, a molded plastic material or a nylon material. The shaft **32** may be axially extending and may be fixed at a central portion of the housing **12**. Connection portions **80**, **82** may be disposed at ends **68**, **70** of the shaft **32**. The connection portions **80**, **82** may be configured to interengage with noncircular connection portions **72** of the housing **12** (e.g., in a central region of the housing **12**).

[0062] In the illustrated embodiment, the connection portions **80**, **82** may include noncircular recesses that may be configured to receive and engage with the noncircular projections **72** of the housing **12**. In another embodiment, the connection portions **80**, **82** may include noncircular projections that may be configured to receive and engage with the noncircular recesses of the housing **12**. The construction of the recesses may be identical, and the construction of the projections may be identical. The projections may have exterior noncircular cross-sections and may be configured to be received within recesses having complementary noncircular interior cross-sections. The noncircular interior and exterior cross-sections cooperate to prevent rotation of the shaft **32** with respect to the housing **12** when the ends of the shaft **32** are mounted interengageably connected with portion of the housing **12** in the assembled rule assembly **10**.

[0063] The end **70** of the shaft **32** may be interiorly threaded to threadedly receive the fastener **138** (as shown in FIG. 5) therein. The fastener **138** may extend through a central hole formed in the second housing member **12b** and may be configured to threadedly engage internal threading in the end **70** of the shaft **32**. The fastener **138** may extend through the connection portion **82** and through the connection portion **72** of the second housing portion **12b**, when the fastener **138** is disposed in the central hole of the second housing member **12b** and the threaded interior end of the shaft **32**. The interengaging connection between the end **70** of the shaft **32** and the first housing member **12b** may be configured to provide a fixed configuration of the shaft **32** at the end **70** of the shaft **32**.

[0064] Similarly, the end **68** of the shaft **32** may be configured to receive a fastener (not shown) therein. The fastener may extend through a central hole formed in the first housing member **12a** and may be configured to be received in the end **68** of the shaft **32**. The fastener may extend through the connection portion **80** and through the connection portion **72** in the first housing member **12a** when the fastener is disposed in the central hole of the first housing member **12a** and the end **68** of the shaft **32**. The fastener through the end **68** of the shaft **32** and the first housing member **12a** may be optional. The interengaging connection between the end **68** of the shaft **32** and the first housing member **12a** may be configured to provide a fixed configuration of the shaft **32** at the end **68** of the shaft **32**.

[0065] The retractable rule assembly **10** may also include a releasable blade lock assembly **84** that is constructed and arranged to be manually actuated to hold the blade **16** in any position of extension outwardly of the housing/blade opening **18** and to release the blade **16** from any position in which it is held. In various embodiments, the releasable blade lock assembly or holding assembly can be of the type described in commonly assigned U.S. Pat. No. 8,117,763, which is hereby incorporated by reference in its entirety.

[0066] Referring to FIGS. **1A-1C** and **10**, the releasable blade lock assembly **84** may include a lock member **86** mounted on the housing **12** for movement in opposite directions between a normally inoperative/release position and a holding/lock position. The lock member **86** may interchangeably referred to as holding member. It can be appreciated that the lock member **86** may have an arcuate member that is movable along an arcuate path between the two positions (i.e., normally inoperative/release position and a holding/lock position). The lock member **86** may have an interior free end portion **88** that is movable into wedging engagement with the tangentially extending

portion of the blade **16** to engage and hold/lock the blade **16** against an interior holding structure **90** on the housing **12** when the lock member **86** is in its lock/holding position. The lock member **86** may have an exterior thumb engaging portion **92** configured to be moved digitally to selectively move the lock member **86** from its normally inoperative/release position and its holding/lock position.

[0067] The lock member **86** is an integral structure made of, for example, an appropriate durable flexible plastic material. The thumb engaging portion **92** is connected by an integral outwardly extending neck portion **94** to an elongated arcuate flexible body portion **96** that terminates in the interior free end **88**. The outwardly extending portion **94** is slidably held within and guided by a slot formed within a front part of housing **12**. The movement of a lower portion of the lock member **86** may be guided by a pair of tabs integrally formed on respective housing members **12a**, **12b**. An integral lock/locking structure **192** of the lock member **86** may be configured to engage a holding structure **194** integrally formed on the housing **12** to releasably lock the lock member **86** in the holding/lock position in wedging engagement with the blade **16**.

[0068] In one embodiment, to lock the blade **16** in a given position of extension, the user (while holding the blade **16** outwardly of the housing **12** against the spring force of the coil spring **22**) slides the thumb engaging portion **92** downwardly with respect to the housing **12** causing the lock structure **192** of the lock member **86** to slide over a ramped surface **198** on the holding structure **194** of the housing **12** and causing the free end **88** of the lock member **86** to move in a locking direction with respect to the blade **16**. The flexible plastic lock structure **192** of the lock member **86** bends resiliently outwardly slightly as it passes over the holding structure **194** of the housing **12**. After the free end **88** contacts the blade **16**, continued movement of the thumb engaging portion **92** in the locking (downward) direction thereafter wedges the free end **88** of the flexible body portion **96** against blade **16** to hold the blade **16** in place against the spring force of the coil spring **22** and moves the lock structure **192** of the lock member **86** into abutting engagement with a locking surface **196** on the holding structure **194**. The lock member **86** may flex slightly as the free end **88** is wedged against the blade **16**. The abutting engagement between the lock structure **192** of the lock member **86** and the locking surface **196** of the housing **12** locks the lock member **86** in its holding/lock position. It can be understood that the blade **16** is held in an extended position (against the spring force of the coil spring **22**) between the free end **88** of the body portion **96** and the interior holding structure **90** by the downward force exerted by the wedged body portion **96**. The interior holding structure **90** may include a series of longitudinally spaced, transversely extending ribs that are constructed and arranged to support the convex side of the blade **16**.

[0069] To release the blade **16**, the user pulls upwardly on the thumb engaging portion **92** which causes the locking structure **192** on the plastic lock member **86** to move resiliently outwardly and past the locking surface **196** of the housing **12** to release the lock member **86** from engagement with the blade **16**. The lock member **86** resiliently returns to its normal arcuate shape.

[0070] When taking a measurement, the user typically holds the housing **12** in one hand and manually pulls the blade **16** out of the housing **12** with the other hand. When a sufficient length of blade **16** has been withdrawn from the housing **12**, the user can lock the blade **16** with respect to the housing **12** using the releasable blade lock assembly **84** to prevent the blade **16** from retracting back into the housing **12** (under the spring force of the coil spring **22**) when the user releases the blade **16**. When the measurement has been taken, the user simply releases the releasable blade lock assembly **84** from locking engagement with the blade **16**. It can be understood that the use of the releasable blade lock assembly **84** when a measurement is being taken is optional. If the releasable blade lock assembly **84** is not used during the taking of a measurement, the user can simply hold the blade **16** with his other hand while the measurement is being taken or, alternatively, the end hook **20** can be placed in hooking engagement with the workpiece to hold the blade **16** outwardly of the housing **12** in a controlled and steady manner against the spring force of the coil spring **22**.

while the measurement is being taken. When the blade **16** is released after taking the measurement, the coil spring **22** rotates the reel **14** with respect to the housing **12** in a blade-winding direction to wind the blade **16** around the reel **14**.

[0071] The blade advancer **24** may include a manually engageable roller **26** (**26.sub.T** or **26.sub.B**). The manually engageable roller **26** may include a top manually engageable roller **26.sub.T** as shown in and described in detail below with respect to FIGS. **1A-4**. The manually engageable roller may include a bottom manually engageable roller **26.sub.B** as shown in and described in detail below with respect to FIGS. **5-9**.

[0072] In one embodiment, the retractable rule assembly **10** may include only the top manually engageable roller **26.sub.T** operating as the blade advancer **24**. The roller **26.sub.T** may be disposed above the blade **16**. In another embodiment, the retractable rule assembly **10** may include only the bottom manually engageable roller **26.sub.B** operating as the blade advancer **24**. The roller **26.sub.B** may be disposed below the blade **16**. In yet another embodiment, the retractable rule assembly **10** may include the top manually engageable roller **26.sub.T** and the bottom manually engageable roller **26.sub.B** both operating as the blade advancer **24**. In yet another embodiment, the manually engageable roller **26** may include a side manually engageable roller disposed on a side of the retractable rule assembly housing (i.e., rather than on the top or the bottom of the retractable rule assembly housing). The side manually engageable roller may have the same structure and configuration as the top manually engageable roller **26.sub.T**. The side manually engageable roller may be disposed on a side of the retractable rule assembly housing that is opposite to the side on which the releasable blade lock assembly **84** is disposed. The side manually engageable roller may be disposed on the same side of the retractable rule assembly housing as the releasable blade lock assembly **84** is disposed. The side manually engageable roller may form the part of the releasable blade lock assembly **84**. The side manually engageable roller may be disposed on a side of the blade **16**. The blade advancer **24** may include any or all of the bottom manually engageable roller **26.sub.B**, the top manually engageable roller **26.sub.T**, or the side manually engageable roller.

[0073] The top manually engageable roller **26.sub.T** is shown in and is described in detail below with respect to FIGS. **1A-4**. The top roller **26.sub.T** may be shown to be disposed near the corner **42** of the housing **12**. In another embodiment, the top roller **26.sub.T** may be disposed near the corner **40** of the housing **12**. In yet another embodiment, the top roller **26.sub.T** may be disposed at any location along the top periphery/circumference of the housing **12**. The top roller **26.sub.T** may be configured to be rotated to extend blade **16** out by engaging the roller **26.sub.T** with the blade **16**. The top roller **26.sub.T** may be referred to as a spring biased roller. The retractable rule assembly **10** may further comprise a spring **126** that biases the roller **26.sub.T** away from the blade **16** and/or reel **14**. The roller **26.sub.T** may be configured to be manually forced against the bias of the spring **106** and into engagement with the blade **16** and/or reel **14** and manually rolled to move the blade **16** in the direction that extends the blade **16** outwardly from the housing **12**. The top roller **26.sub.T** may be disposed above the blade **16** and is normally held away from the blade due to the bias of the spring **106**.

[0074] The top roller **26.sub.T** may have a first end portion **108** and an opposing second end portion **110**. The top roller **26.sub.T** may include a cylindrical/peripheral side wall **112** extending between the first end portion **108** and the second end portion **110**. The top roller **26.sub.T** may include end walls **114**, **116** disposed at the first end portion **108** and the second end portion **110**. The end walls **114**, **116** may be perpendicular to the cylindrical side wall **112**. The top roller **26.sub.T** may include protrusions **118** disposed centrally (i.e., in a central portion CP") in the end walls **114**, **116**. The protrusions **118** may extend outwardly away from the respective end walls **114**, **116**. The protrusions **118** may extend in an axial direction RA-RA and may extend to be parallel to the cylindrical side wall **112**. The top roller **26.sub.T** may be made of one or more of a plastic, rubber, and metal material. This roller can be single or multi-material.

[0075] The first housing portion **12a** and the second housing portion **12b** may each include a roller

receiving portion **120** that is configured to receive the top roller **26.sub.T** therein such that at least a portion of the top roller **26.sub.T** may be protruding slightly outwardly and away from the housing **12** and at least a portion of the top roller **26.sub.T** may be disposed in the housing **12**. A surface/portion **134** of the cylindrical side wall **112** of at least a portion of the top roller **26.sub.T** protruding slightly outwardly and away from the housing **12** may form a manually engageable surface/portion **134** of the top roller **26.sub.T**. In another embodiment, the surface/portion **134** of the cylindrical side wall **112** of at least a portion of the top roller **26.sub.T** may not protrude slightly outwardly and away from the housing **12** and the housing **12** may have an opening through which a user may engage the surface/portion **134** of the cylindrical side wall **112** of at least a portion of the top roller **26.sub.T**. That is, a user engaging the manually engageable surface/portion **134** of the top roller **26.sub.T** may manually force the roller **26.sub.T** against the bias of the spring **106** and into engagement with the blade **16** and/or reel **14** and manually roll the top roller **26.sub.T**. This action may move the blade **16** in the direction that extends the blade **16** outwardly from the housing **12**. A surface/portion **136** of the cylindrical side wall **112** of at least a portion of the top roller **26.sub.T** disposed in the housing **12** may form the blade and/or reel engageable surface/portion **136** of the top roller **26.sub.T**. The surfaces/portions **134** and **136** of the roller **26.sub.T** may form opposing surfaces/portions of the top roller **26.sub.T**. That is, the surface/portion **136** of the cylindrical side wall **112** of at least a portion of the top roller **26.sub.T** disposed in the housing **12** may be forced/moved into engagement with the blade **16** and/or reel **14** to move the blade **16** in the direction AD (as shown in FIGS. **10A-12D**) that extends the blade **16** outwardly from the housing **12**. As the top roller **26.sub.T** may be (manually) rolled/rotated/moved, the portions of the top roller **26.sub.T** that form surfaces/portions **134**, **136** may change. In general, the surface/portion **136** of the cylindrical side wall **112** of at least a portion of the roller **26.sub.T** may be disposed in the housing **12**, while the surface/portion **134** of the cylindrical side wall **112** of at least a portion of the roller **26.sub.T** may be protruding slightly outwardly and away from the housing **12**.

[0076] The spring **126** may be one of the two springs **126.sub.a** and **126.sub.b** of the retractable rule assembly **10**. The first housing portion **12a** and the second housing portion **12b** may include spring receiving portions **124** that are configured to receive the associated springs **126.sub.a** and **126.sub.b** therein. Each spring receiving portion **124** may include a cylindrical shaped opening to receive the corresponding spring **126.sub.a**, **126.sub.b** therein. Each spring **126.sub.a**, **126.sub.b** may have a compressed state/configuration in which the roller **26.sub.T** may be configured to be manually forced against the bias of the spring **106** and into engagement with the blade **16** and/or reel **14**, and an expanded state/configuration (i.e., the normal state/configuration of the spring **126.sub.a**, **126.sub.b**) in which the spring **126.sub.a**, **126.sub.b** may be configured to bias the roller **26.sub.T** away from the blade **16** and/or reel **14**. Ends **130** of the springs **126.sub.a**, **126.sub.b** may be disposed on a surface **128** of the spring receiving portions **124** and opposing ends **132** of the spring **126.sub.a**, **126.sub.b** (in the normal state/configuration of the spring **126.sub.a**, **126.sub.b**) may be in contact with or engagement with the corresponding protrusions **118** of the roller **26.sub.T** to bias the roller **26.sub.T** away from the blade **16** and/or reel **14**.

[0077] The bottom manually engageable roller **26.sub.B** is shown in and described in detail below with respect to FIGS. **5-9**. The bottom roller **26.sub.B** may be disposed below the blade **16**. The bottom roller **26.sub.B** may normally be held away from the blade **16** and/or the reel **14** under the force of gravity. The bottom roller **26.sub.B** may include a first position (e.g., a normal position/configuration) in which the roller **26.sub.B** may be held away from the blade **16** and/or the reel **14** under the force of gravity and a second position in which the bottom roller **26.sub.B** may be manually forced against the force of gravity and into engagement with the blade **16** and/or reel **14** and manually rolled to move the blade **16** in the direction that extends the blade **16** outwardly from the housing **12**. The bottom roller **26.sub.B** may be referred to as a heely tape or a gravity roller. That is, the return force (to its normal position/configuration) for the bottom roller **26.sub.B** may be provided by the force of gravity rather than springs (as is the case with the top roller **26.sub.T**).

[0078] The corner portion **36** of the second housing portion **12b** may be configured to include a bottom roller support portion **140**. The bottom roller support portion **140** may include a protrusion **142**, an opening **144** extending through the protrusion **142**, and peripheral/circumferential support portions **146** extending outwardly from the protrusion **142**. In the illustrated embodiment, four peripheral/circumferential support portions **146** are shown to be extending outwardly from the protrusion **142**. The number of circumferential support portions **146** may vary. In another embodiment, three circumferential support portions **146** may be extending outwardly from the protrusion **142**.

[0079] The bottom roller **26.sub.B** may be configured to be rotated to extend blade **16** out by engaging the bottom roller **26.sub.B** with the blade **16**/the reel **14**. The bottom roller **26.sub.B** may include a first end portion **156**, a second end portion **158**, and an opening **148** therein. The opening **148** may be disposed centrally (in a central portion) of the bottom roller **26.sub.B** and may be extending between the first end portion **156** and the second end portion **158** of the bottom roller **26.sub.B**. The bottom roller **26.sub.B** may also include a cylindrical/peripheral side wall **160** extending between the first end portion **156** and the second end portion **158**.

[0080] When the bottom roller **26.sub.B** is disposed in the housing **12b**, inner circumferential surfaces **150** of the opening **148** of the bottom roller **26.sub.B** may be surround outer circumferential surfaces **152** of the protrusion **142**. Also, when the bottom roller **26.sub.B** is disposed in the housing **12b**, end surfaces **147** (facing the second housing portion **12b** and disposed at the second end portion **158**) of the bottom roller **26.sub.B** may be supported by surfaces **154** of the circumferential support portions **146**. These configurations of the bottom roller **26.sub.B** and the bottom roller support portion **140** enable the bottom roller **26.sub.B** be supported by the bottom roller support portion **140**. A clearance/gap may be provided between the outer circumferential surfaces **152** of the protrusion **142** and the inner circumferential surfaces **150** of the opening **148** of the bottom roller **26.sub.B**. This clearance/gap may enable the movement of the bottom roller **26.sub.B** between its first and its second positions. A clearance/gap may also be provided between the end surfaces **147** (facing the second housing portion **12b** and disposed at the second end portion **158**) of the bottom roller **26.sub.B** and the surfaces **154** of the circumferential support portions **146** to enable the movement of the bottom roller **26.sub.B** between its first and its second positions. As shown in FIG. **8**, the outer circumferential surfaces **152** of the protrusion **142** and the inner circumferential surfaces **150** of the opening **148** of the roller **26.sub.B** may have stepped configurations and may be complementary to each other.

[0081] The opening **144** of the housing member **12b** may be configured to align with an opening **145** of the other of the housing member **12a**. The fastener **44** may then be positioned in aligned openings **144**, **145** of the corner portion **36** of the housing member **12a**, **12b**. Thus, it can be appreciated that the housing members **12a**, **12b** may be secured together using fastener in this location. The fastener **44** may also enable the roller **26.sub.B** be supported by the bottom roller support portion **140** and be secured in the housing **12**.

[0082] The housing **12** may be configured to receive the bottom roller **26.sub.B** therein such that at least a portion of the bottom roller **26.sub.B** may be protruding slightly outwardly and away from the housing **12** and at least a portion of the bottom roller **26.sub.B** may be disposed in the housing **12**. A surface/portion **162** of the cylindrical side wall **160** of at least a portion of the bottom roller **26.sub.B** protruding slightly outwardly and away from the housing **12** may form a manually engageable surface/portion **162** of the bottom roller **26.sub.B**. In another embodiment, the surface/portion **162** of the cylindrical side wall **160** of at least a portion of the bottom roller **26.sub.B** may not protrude slightly outwardly and away from the housing **12** and the housing **12** may have an opening through which a user may engage the surface/portion **162** of the cylindrical side wall **160** of at least a portion of the bottom roller **26.sub.B**. That is, a user engaging the manually engageable surface/portion **162** of the bottom roller **26.sub.B** may manually force the bottom roller **26.sub.B** against the force of gravity and into engagement with the blade **16** and/or

reel **14** and manually roll the bottom roller **26.sub.B**. This action may move the blade **16** in the direction that extends the blade **16** outwardly from the housing **12**. A surface/portion **164** of the cylindrical side wall **160** of at least a portion of the bottom roller **26.sub.B** disposed in the housing **12** may form the blade and/or reel engageable surface/portion **164** of the bottom roller **26.sub.B**. That is, the surface/portion **164** of the cylindrical side wall **160** of at least a portion of the bottom roller **26.sub.B** disposed in the housing **12** may be forced/moved into engagement with the blade **16** and/or reel **14** to move the blade **16** in the direction that extends the blade **16** outwardly from the housing **12**.

[0083] The surfaces/portions **162** and **164** of the bottom roller **26.sub.B** may be disposed at an angle (between 90 and 180 degrees) with respect to each other. In another embodiment, the bottom roller **26.sub.B** may be disposed anywhere along the bottom of the retractable rule assembly **10**. In such an embodiment, the surfaces/portions **162** and **164** of the bottom roller **26.sub.B** may form opposing surfaces of the bottom roller **26.sub.B**. As the bottom roller **26.sub.B** may be manually rolled/rotated/moved, portions of the bottom roller **26.sub.B** that form the surfaces/portions **162**, **164** may change. In general, the surface/portion **164** of the cylindrical side wall **160** of at least a portion of the bottom roller **26.sub.B** may be disposed in the housing **12**, while the surface/portion **162** of the cylindrical side wall **160** of at least a portion of the bottom roller **26.sub.B** may be slightly protruding outwardly and away from the housing **12**.

[0084] The blade advancer **24** may form part of the releasable blade lock assembly **84** as shown in FIGS. **10A-12D**. The blade lock assembly **84** may be constructed and arranged to be manually actuated to lock the blade **16** in any position of extension outwardly of the housing opening **18** and to release the blade **16** from any position in which the blade **16** is locked. The structure and operation of the blade lock assembly **84** is explained in detail above. The blade lock assembly **84** may be movable in a first direction **FD** to lock the blade **16** and a second direction **SD**, opposite direction, to advance the blade **16** outwardly from the housing **12**. For example, the blade advancer **24** may include a dual action slidelock is shown in and described in detail below with respect to FIGS. **10A-12D**. The blade advancer **24** of FIGS. **10A-12D** may have the same ergonomics as the blade lock assembly **84**.

[0085] FIGS. **10A-10D** show various views of the retractable rule assembly **10** where the blade advancer **24** may form part of the releasable blade lock assembly **84**. The releasable blade lock assembly **84** is shown in a locked configuration in FIG. **10A** and an unlocked configuration in FIG. **10B**. The blade advancer **24** is shown in a spool/reel engaged configuration in FIG. **10C** and in a blade extended configuration in FIG. **10D** in which the blade **16** is advanced in the direction **AD** that extends the blade **16** outwardly from the housing **12**.

[0086] Referring to FIGS. **10A-10D**, the blade advancer **24** may include a first engaging portion **166** disposed on the blade lock assembly **84** and a second engaging portion **168** disposed on the reel **14**. The first engaging portion **166** of the blade lock assembly **84** may be configured to interlock/interengage with the second engaging portion **168** of the reel **14** to facilitate the advancement/movement of the blade **16** in the direction **AD** (i.e., outwardly from the housing **12**). The first engaging portion **166** may be in the form of a projection or a tooth **170**. The first engaging portion **166** may interchangeably referred to as spool/reel interaction portion/tooth. The second engaging portion **168** may include teeth **172** that are configured to engage and interlock with the first engaging portion **166**. The reel **14** may be modified to include the second engaging portion **168** including the teeth **172**. The housing **12** may be modified to include a track for the blade lock assembly **84** to be slid/moved/rotated back in the reverse direction (i.e., the second direction **SD**). The modified track may enable the blade lock assembly **84** move/rotate/slide, in the second direction **SD**, past the unlock configuration/position. Although the first engaging portion **166** may be shown to include the projection/tooth **170** and the second engaging portion **168** may be shown to include the teeth **172**, in other embodiments, the first engaging portion **166** and the second engaging portion **168** may include other interlocking/interengaging portions that are configured to

enable the advancement/movement of the blade **16**, in the direction AD, outwardly from the housing **12**.

[0087] FIGS. **10A-10B** show a lock and an unlock configurations of the blade lock assembly **84**. The blade lock assembly **84** may be movable in the first direction FD to lock the blade **16** in any position of extension outwardly of the housing opening **18** and the second direction SD, opposite direction, to unlock the blade **16** from any position in which the blade **16** is locked. The lock and unlock configurations of the blade lock assembly **84** are explained in detail above.

[0088] To advance/move the blade **16** in the direction AD (i.e., outwardly from the housing **12**), the releasable blade lock assembly **84** may be first slid/moved/rotated in the second direction SD. For example, the releasable blade lock assembly **84** may be slid/moved/rotated in the second direction SD from its lock configuration to its unlock configuration. As noted above, the second direction SD of the releasable blade lock assembly **84** may be opposite to the first direction FD of the releasable blade lock assembly **84**. In the unlock configuration in FIG. **10B**, the releasable blade lock assembly **84** may then be pushed into the housing **12** (i.e., in the inward radial direction as shown by the arrow ID). This pushing action may cause the first engaging portion **166** (e.g., the projection/tooth **170**) of the releasable blade lock assembly **84** to interlock with the second engaging portion **168** (e.g., teeth **172**) on the rim of the reel **14**. The interlock engagement between the first engaging portion **166** (e.g., the projection/tooth **170**) of the releasable blade lock assembly **84** and the second engaging portion **168** (e.g., teeth **172**) on the rim of the reel **14** enables the reel **14** and the blade **16** to be moved along with the releasable blade lock assembly **84**. In this interlocked (or spool/reel engaged) configuration as shown in FIG. **10C**, and while maintaining (radial) pressure on the releasable blade lock assembly **84**, the releasable blade lock assembly **84** may further be moved/slid/rotated in the second direction SD (from the unlock configuration of the blade lock assembly **84**) to advance/move the blade **16** in the direction AD to extend the blade **16** outwardly from the housing **12**. Once the blade **16** is moved/advanced in the direction AD to extend the blade **16** outwardly from the housing **12**, the user may grab the end hook **20** and extend the blade **16** further outwardly from the housing **12** so as to engage a portion of the end hook **20** with a workpiece, or may engage a workpiece directly with the end hook **20** and allow the workpiece to anchor the blade **16** to permit one handed operation by a user.

[0089] The user may move/pull the releasable blade lock assembly **84** in an outward radial direction (i.e., opposite to the direction of the arrow ID) to release/disengage the interlock connection between the first engaging portion **166** (e.g., the projection/tooth **170**) of the releasable blade lock assembly **84** and the second engaging portion **168** (e.g., teeth **172**) on the rim of the reel **14**. When there is an interlock engagement between the first engaging portion **166** (e.g., the projection/tooth **170**) of the releasable blade lock assembly **84** and the second engaging portion **168** (e.g., teeth **172**) on the rim of the reel **14** and when the releasable blade lock assembly **84** is moved in the second direction SD, the releasable blade lock assembly **84** may act as the blade advancer. When the interlock engagement between the first engaging portion **166** (e.g., the projection/tooth **170**) of the releasable blade lock assembly **84** and the second engaging portion **168** (e.g., teeth **172**) on the rim of the reel **14** is released/disengaged, the releasable blade lock assembly **84** may act as the releasable blade lock. That is, the releasable blade lock assembly **84** may then be moved back in the first direction FD to lock the blade **16** in any position of extension outwardly of the housing opening **18**. The releasable blade lock assembly **84** may also be moved back in the first direction FD to the home position of the blade advancer **24** when the blade advancer's function is desired. The home position of the blade advancer **24** may include the unlock configuration/position of the releasable blade lock assembly **84**.

[0090] In one embodiment, the blade advancer **24** of FIGS. **10A-10D** may be configured to be spring-biased to return the releasable blade lock assembly **84** automatically from its interlock engagement configuration (i.e., in which the projection/tooth **170** of the releasable blade lock assembly **84** and the teeth **172** of the second engaging portion **168** on the rim of the reel **14** are in

an interlock engagement) to a releasable blade lock configuration (i.e., in which the projection/tooth **170** of the releasable blade lock assembly **84** is released/disengaged from the teeth **172** of the second engaging portion **168** on the rim of the reel **14**).

[0091] FIGS. **11A-11D** show various views of the retractable rule assembly **10** where the blade advancer **24** may form part of the releasable blade lock assembly **84**. The releasable blade lock assembly **84** is shown in a locked configuration in FIG. **11A** and an unlocked configuration in FIG. **11B**. The blade advancer **24** is shown in a friction pad engaged configuration in FIG. **11C** and in a blade extended configuration in FIG. **11D** in which the blade **16** is advanced in the direction AD that extends the blade **16** outwardly from the housing **12**. The embodiment shown in FIGS. **11A-11D** is similar to that described in detail in FIGS. **10A-10D** except for the differences noted below. The embodiment of the retractable rule assembly **10** as shown in FIGS. **11A-11D** does not have any modifications to the spool/reel **14**.

[0092] Referring to FIGS. **11A-11D**, the blade advancer **24**, which forms part of releasable blade lock assembly **84**, may include a cam surface. The cam surface may be configured to force a friction pad (rubber push) feature **176** against the blade **16**/reel **14**.

[0093] To advance/move the blade **16** in the direction AD (i.e., outwardly from the housing **12**), the releasable blade lock assembly **84** may be first slid/moved/rotated in the second direction SD. For example, the releasable blade lock assembly **84** may be slid/moved/rotated in the second direction SD from its lock configuration to its unlock configuration. As noted above, the second direction SD the releasable blade lock assembly **84** may be opposite to the first direction FD of the releasable blade lock assembly **84**. In the unlock configuration in FIG. **11B**, the releasable blade lock assembly **84** may then be pushed into the housing **12** (i.e., in the inward radial direction as shown by the arrow ID). This pushing action may cause the cam surface to force the friction pad (rubber push) feature **176** against the blade **16**/reel **14**. In this friction pad engaged configuration as shown in FIG. **11C**, and while maintaining (radial) pressure on the releasable blade lock assembly **84**, the releasable blade lock assembly **84** may further be moved/slid/rotated in the second direction SD (from the unlock configuration of the blade lock assembly **84**) to advance/move the blade **16** in the direction AD to extend the blade **16** outwardly from the housing **12**. Once the blade **16** is moved/advanced in the direction AD to extend the blade **16** outwardly from the housing **12**, the user may grab the end hook **20** and may extend the blade **16** further outwardly from the housing **12** so as to engage a portion of the end hook **20** with a workpiece, or may engage a workpiece directly with the end hook **20** and allow the workpiece to anchor the blade **16** to permit one handed operation by a user. The blade advancer **24** of FIGS. **11A-11D** may also include a detent in the cam surface that indicates a maximum extension of the blade advancer **24**/the blade lock assembly **84** (in the second direction SD) to a user. After the user hooks the blade **16** on the workpiece, the user may pull the blade **16** to extend the blade **16**. This may cause the blade advancer **24**/the blade lock assembly **84** to travel past the detent, where the cam surface releases the friction pad feature from the blade **16**/reel **14**. The releasable blade lock assembly **84** may also be moved back in the first direction FD to the home position of the blade advancer **24** when the blade advancer's function is desired. The home position of the blade advancer **24** may include the unlock configuration/position of the releasable blade lock assembly **84**.

[0094] In one embodiment, the blade advancer **24** of FIGS. **11A-11D** may be configured to be spring-biased to return the releasable blade lock assembly **84**/the blade advancer **24** automatically from its friction pad engaged configuration (i.e., as shown in FIGS. **11C** and **11D**) to its releasable blade lock configuration (in which its friction pad is disengaged from the rim of the reel **14**, for example, as shown in FIGS. **11A** and **11B**).

[0095] It may be appreciated that a cam surface "track" could drive the friction pad against the blade **16** and/or reel **14**. In such a configuration, the user may not need to exert as much downward force (radial) as the lock assembly **84** is pulled back. This being said, in some embodiments, the user must exert some radial force at the start of the extend operation, otherwise the cam follower

could travel in the return path.

[0096] Referring to the embodiment in FIGS. **10A-11D**, the “braking” feature is added to the blade lock assembly **84**, this feature enables engaging of the spool and may work similar to a finger brake (thumb brake). In some embodiments, the braking feature may simply push radially on blade lock assembly **84** so that the friction pad drags on the spool/tape. As the tape is retracting via the spring, the user can push down on the button and slow or stop retraction, depending on the radial force used.

[0097] FIGS. **12A-12D** show various views of the retractable rule assembly **10** where the blade advancer **24** may form part of the releasable blade lock assembly **84**. The releasable blade lock assembly **84** is shown in a locked configuration in FIG. **12A** and an unlocked configuration in FIG. **12B**. The blade advancer **24** is shown in a releasable blade lock configuration in FIG. **12C** and in a blade extended configuration in FIG. **12D** in which the blade **16** is advanced in the direction AD that extends the blade **16** outwardly from the housing **12**. The embodiment shown in FIGS. **12A-12D** is similar to that described in detail in FIGS. **10A-10D** and FIGS. **11A-11D** except for the differences noted below. The embodiment shown in FIGS. **12A-12D** does not have any modifications to the spool/reel **14**.

[0098] Referring to FIGS. **12A-12D**, the blade advancer **24**, which forms part of releasable blade lock assembly **84**, may include an end hook pusher **178**. The end hook pusher **178** may be formed from spring steel and be a spring steel pusher or a spring steel member. In other embodiments the end hook pusher **178** may be formed from plastic or so on. In some embodiment the end hook pusher **178** may be formed as an assembly, such as from links of chain, or any other appropriate collectively flexible member (e.g., with rigidity when pushed, yet flexibility when pulled). The end hook pusher **178** may be configured to engage with the end hook **20** (i.e., connected to the distal end **28** of the blade **16**) to push the end hook **20** (and the blade **16** connected thereto) outwardly from the housing **12**, that is in the direction AD. It may be appreciated that in some embodiments the user may push in a button of the blade lock assembly **84** to get past the unlocked position, before sliding the blade lock assembly **84** back. This extends the end hook pusher **178** which in turn pushes the hook and extends the blade against the spring bias. User can then hook on a workpiece single-handedly or pull on the blade to extend it further. The pusher could retract via the spring bias or manually in various embodiments.

[0099] To advance/move the blade **16** in the direction AD (i.e., outwardly from the housing **12**), the releasable blade lock assembly **84** may be first slid/moved/rotated in the second direction SD. For example, the releasable blade lock assembly **84** may be slid/moved/rotated in the second direction SD from its lock configuration to its unlock configuration. As noted above, the second direction SD the releasable blade lock assembly **84** may be opposite to the first direction FD of the releasable blade lock assembly **84**. Comparing the lock configuration in FIG. **12A** and the unlock configuration of FIG. **12B** shows that the sliding/movement/rotation of the releasable blade lock assembly **84** in the second direction SD causes the movement of the end hook pusher **178**. FIG. **12B** shows the end hook pusher **178** engaging with the end hook **20** of the blade **16**. In the unlock configuration in FIG. **12B**, the releasable blade lock assembly **84** may then be pushed into the housing **12** (i.e., in the inward radial direction as shown by the arrow ID). This pushing action allows movement of the end hook pusher **178** to rotate and engage with the end hook **20** of the blade **16**. In the configuration as shown in FIG. **12C**, and while maintaining (radial) pressure on the releasable blade lock assembly **84**, the releasable blade lock assembly **84** may further be moved/slid/rotated in the second direction SD (from the unlock configuration of the blade lock assembly **84**) to advance/move the blade **16** in the direction AD to extend the blade **16** outwardly from the housing **12**. Once the blade **16** is moved/advanced in the direction AD to extend the blade **16** outwardly from the housing **12**, the user may grab the end hook **20** and may extend the blade **16** further outwardly from the housing **12** so as to engage a portion of the end hook **20** with a workpiece, or may engage a workpiece directly with the end hook **20** and allow the workpiece to

anchor the blade **16** to permit one handed operation by a user. After the user hooks the blade **16** on the workpiece, the user may pull the blade **16** to extend the blade **16**. The releasable blade lock assembly **84** may also be moved back in the first direction FD to the home position of the blade advancer **24** when the blade advancer's function is desired. The home position of the blade advancer **24** may include the unlock configuration/position of the releasable blade lock assembly **84**.

[0100] In one embodiment, the blade advancer **24** of FIGS. **12A-12D** may be configured to be spring-biased to return the releasable blade lock assembly **84**/the blade advancer **24** automatically from its configuration (i.e., as shown in FIGS. **12C** and **12D**) to its releasable blade lock configuration (as shown in FIG. **12B**).

[0101] The blade advancer **24** may be a spring biased blade advancer **26.sub.SB**. For example, the blade advancer **26.sub.SB** may include a microSD (i.e., a type of removable flash memory card used for storing information) type configuration (as shown in and described in detail below with respect to FIGS. **13-15**). The spring biased blade advancer **26.sub.SB** may comprise a pusher **180** and a spring **182**. The blade advancer **26.sub.SB** may be constructed and arranged such that when the blade advancer **26.sub.SB** is actuated, the spring **182** pushes the pusher **180** to engage a portion of the blade **16**, the end hook **20** (though some may interpret the blade **16** as including the end hook **20**), or the reel **14** against a bias of the coil spring **22** to advance the blade **16** in the direction AD that extends the blade **16** outwardly from the housing **12**. The pusher **180**, when moved by the spring **182**, contacts the end hook **20** to extend the blade **16** outwardly from the housing **12** that is in the direction AD.

[0102] The pusher **180** may interchangeably be referred to as a blade feeder. The pusher **180** may be configured to support a portion of the blade **16** thereon. The pusher **180** may be constructed of, for example, a molded plastic material. The pusher **180** may be configured to be movable with respect to the housing **12**. The pusher **180** may be configured to be movably connected to a portion of the housing **12**. The pusher **180** may be operatively connected to the spring **182**. When the blade advancer **26.sub.SB** is actuated, the spring **182** may be configured to push the pusher **180** (e.g., in the direction AD) to engage a portion of the blade **16**/the reel **14** against the bias of the coil spring **22** to advance the blade **16** in the direction AD that extends the blade **16** outwardly from the housing **12**.

[0103] The spring **182** may interchangeably be referred to as an extension/a tension spring in some nonlimiting embodiments (i.e., the spring is designed to operate with a tension load, so the spring stretches as the load is applied to it). The extension spring **182** may be constructed and arranged to become extended under load. The turns/loops/coils of the extension spring **182** may normally be touching each other in the unloaded position/configuration. The spring **182** may have a hook, eye or some other means of attachment at each end. One end of the spring **182** may be connected to a portion of the housing **12** and the other end of the spring **182** may be connected to a portion of the pusher **180**. It may be appreciated, however, that in various embodiments the spring can be extended, compressed, or twisted to generate a spring force that may power popping out the blade **16** when the pusher **180** is pressed into the housing **12**. It may further be appreciated that the blade **16** may extend from the housing **12** by a distance that is greater than (e.g., a multiple of) the insertion distance that deforms the spring.

[0104] A spring force of the blade advancer spring **182** may be sufficiently large to overcome the spring force of the coil spring **22** retracting the blade **16** into the housing **12**. The blade advancer **26.sub.SB** may remain extended until manually forced to a retracted position. The blade advancer **26.sub.SB** may also serve as a shock absorber against rapid retraction of the blade **16** while in the extended position. The blade advancer **26.sub.SB** may also serve as additional finger support while in the extended position. To retract the pusher **180** of the blade advancer **26.sub.SB** into the housing **12**, the pusher **180** may simply be pressed back into the housing **12**. The blade advancer **26.sub.SB** may be configured to provide a pop out blade presentation and to provide soft close (retracting the

pusher **180** of the blade advancer **26.sub.SB** into the housing **12**) configuration. The blade advancer **26.sub.SB** may include a push-push style ejection system (or push-push type connector). For the push-push type style ejection system, the blade **16**/the end hook **20** may be inserted into the housing **12** and may be extracted from housing **12** by pushing the pusher **180**/the blade **16**/the end hook **20** in an insertion direction (that is opposite to the direction AD). The pusher **180** may be pushed into the housing **12** by approximately $\frac{1}{4}$ inch, this may cause the blade **16** to pop out by approximately $1\frac{1}{4}$ inch.

[0105] In another embodiment, the blade advancer **26.sub.SB** may include a push-pull style ejection system (or push-pull type connector). For the push-pull style ejection system, the blade **16** may be extracted from the housing **12** by applying a force to the pusher **180**/the blade **16**/the end hook **20** in a direction opposite to the insertion direction.

[0106] The present patent application provides an auto-extension tape measure **10** as shown in and described in detail with respect to FIGS. **16-17**. FIGS. **16** and **17** show a retractable rule assembly or tape measure **10** that may be configured to automatically extend the blade **16** with a press of a button or the engagement of a lever **184**.

[0107] A mechanism, both internal and external to the tape measure **10**, may engage with the internal tape measure components to extend the blade **16** out of the tape measure **10**. The mechanism may be made operational by actuation of a user/manually engageable feature (e.g., lever **184** or a button) on the exterior of the tape measure **10**. As will be clear from the discussions below, the mechanical engagement of the manually engageable feature may interact with the internal components of the tape measure **10**, forcing the blade **16** out of the mouth/blade opening **18** of the tape measure **10** to a distance set by the amount of engagement of the manually engageable feature. The release of this mechanism may return the components of the tape measure **10** to the rest position allowing the internal components of the tape measure **10** to move freely.

[0108] FIG. **16** shows the lever **184** in its rest position/configuration, while FIG. **17** shows the lever **184** in its operational position/configuration. A downward force (e.g., in the direction of an arrow DF) slides the lever **185** (from its rest position/configuration) along the slot **186** to its operational position/configuration. This causes an engagement between a drive gear **188** and a driven gear **190**. The present patent application provides a tape rule pop out blade **16** where an external mechanism (e.g., the lever **184**) translates a push action into rotation of the tape reel **14**, so as to ease grabbing the end hook **20** and the blade **16** for further extension. For example, the spring-loaded lever **184**, external to the tape measure **10**, may manually engaged by the user. Upon engagement of the lever **184**, the lever assembly may slide along the slot **186** designed into the tape measure case/housing **12**. The lever **184** may be fixed to the gear **188** internal to the tape measure **10**. At the bottom of the slot **186**, the internal gear **188** that is fixed to the spring-loaded lever **184** may become engaged with a secondary gear **190** that is fixed to the cartridge/reel **14** of the tape measure **10**. The secondary gear **190** may interchangeably referred to as the driven gear and the internal gear **188** may interchangeably referred to as the drive gear. Upon rotation of the gear **190**, activated by the lever **184**, the cartridge/reel **14** may rotate that extends the blade **16** outwards. Upon release of the lever **184**, a spring activated linkage internal to the tape measure case/housing **12** may pull the lever **184** and the gear assembly back to its rest position, allowing the cartridge/reel **14** to rotate freely.

[0109] The present patent application may use a lever, a pushbutton, a gear, or a friction wheel, etc. to effectuate the movement of the reel **14**, but the present patent application may use any other mechanism (e.g., an external mechanism) on a spring-retractable tape measure **10** to translate a push action into rotation of the tape reel **14** so as to ease grabbing the end hook **20** and the blade **16** for further extension.

[0110] At least some portions (e.g., where the user's palm, fingers and/or thumb may generally be in overlying relation with the housing **12**) of the housing portions **12a** and **12b** may be covered with, for example, the overmolded rubber material or a rubber-like polymeric material. For

example, the housing portions **12a** and **12b** may have outer user engageable portions/surfaces (or gripping areas) that may be covered with, for example, the overmolded rubber material or a rubber-like polymeric material.

[0111] A bottom wall of the housing **12** may provide a finger grip enhancing configuration for a gripping hand of the user. The bottom wall may have a forward end portion disposed adjacent to the blade opening **18** and a rearward end portion at the opposite end of the bottom wall. A portion of the bottom wall (i.e., between the forward and the rearward end portions) may be generally recessed to provide the finger grip enhancing configuration for the gripping hand of the user. The gripping area on the bottom wall of the housing **12** may be covered with, for example, the overmolded rubber material or a rubber-like polymeric material. It can thus be appreciated that the housing **12** is constructed to be easily held in one hand of a user such that the user's fingers may engage the finger grip enhancing portion and the user's palm and thumb may generally be in overlying relation with a top portion of the housing **12**. A peripheral portion of housing **12** may also be provided with a coating around gripped portions of the housing **12** to provide increased frictional engagement between the housing **12** and a user's hand and to provide a relatively soft comfortable surface for the user's hand. The coating may be made from, for example, a rubber-like material.

[0112] Example embodiments have been provided so that this disclosure will be thorough, and to fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

[0113] In one embodiment, dimensions/numerical ratios or ranges of the present patent application may be up to 5 percent, 10 percent, 15 percent or 20 percent greater than or up to 5 percent, 10 percent, 15 percent or 20 percent less than the values described throughout the present patent application. In another embodiment, dimensions/numerical ratios or ranges of the present patent application may be in the range of ± 5 percent or ± 10 percent of the values described throughout the present patent application.

[0114] The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

[0115] When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0116] Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

[0117] Terms of degree such as “generally,” “substantially,” “approximately,” and “about” may be used herein when describing the relative positions, sizes, dimensions, or values of various elements, components, regions, layers and/or sections. These terms mean that such relative positions, sizes, dimensions, or values are within the defined range or comparison (e.g., equal or close to equal) with sufficient precision as would be understood by one of ordinary skill in the art in the context of the various elements, components, regions, layers and/or sections being described.

[0118] Numerous modifications may be made to the exemplary implementations described above. These and other implementations are within the scope of this application.

[0119] Although the present patent application has been described in detail for the purpose of illustration, it is to be understood that such detail is solely for that purpose and that the present patent application is not limited to the disclosed embodiments, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. In addition, it is to be understood that the present patent application contemplates that, to the extent possible, one or more features of any embodiment can be combined with one or more features of any other embodiment.

Claims

1. A retractable rule assembly comprising: a housing; a reel rotatably mounted in the housing; an elongated blade wound on the reel and constructed and arranged to extend outwardly through an opening in the housing; a hook positioned on a distal end of the blade; a coil spring configured to rotate the reel in the housing assembly in a direction to wind up the elongated blade when extending outwardly of the housing; and a blade advancer constructed and arranged to be manually engaged and moved relative to the housing and into contact with the blade and/or the reel, the blade advancer being movable against a bias of the coil spring to advance the blade in a direction that extends the blade outwardly from the housing.
2. The retractable rule assembly of claim 1, wherein the blade advancer comprises a manually engageable roller.
3. The retractable rule assembly of claim 2, further comprising a spring that biases the roller away from the blade and/or reel, and wherein the roller is configured be manually forced against the bias of the spring and into engagement with the blade and/or reel and manually rolled to move the blade in the direction that extends the blade outwardly from the housing.
4. The retractable rule assembly of claim 2, the roller is disposed above the blade and is normally held away from the blade.
5. The retractable rule assembly of claim 2, wherein the roller is disposed below the blade and is normally held away from the blade under the force of gravity.
6. The retractable rule assembly of claim 1, wherein the blade advancer forms part of a releasable blade lock assembly, the blade lock assembly constructed and arranged to be manually actuated to lock the blade in any position of extension outwardly of the housing opening and to release the blade from any position in which it is locked.
7. The retractable rule assembly of claim 6, wherein the blade lock assembly is movable in a first direction to lock the blade and a second direction, opposite direction, to advance the blade.

8. The retractable rule assembly of claim 7, wherein the blade advancer includes a first engaging portion disposed on the blade lock assembly and a second engaging portion disposed on the reel, wherein the first engaging portion of the blade lock assembly is configured to interlock with the second engaging portion of the reel to facilitate advancement of the blade in the direction that extends the blade outwardly from the housing, and wherein the first engaging portion includes a projection and the second engaging portion includes teeth that are configured to engage and interlock with the projection.

9. The retractable rule assembly of claim 7, wherein the blade advancer, which forms part of releasable blade lock assembly, includes a cam surface, and wherein the cam surface is configured to force a friction pad feature against the blade/reel to facilitate advancement of the blade in the direction that extends the blade outwardly from the housing.

10. The retractable rule assembly of claim 7, wherein the blade advancer, which forms part of releasable blade lock assembly, includes an end hook pusher, wherein the end hook pusher is configured to engage with the hook positioned on the distal end of the blade to push the hook and the blade connected thereto outwardly from the housing.

11. The retractable rule assembly of claim 1, wherein the blade advancer includes a manually engageable portion disposed on the housing and moveable along a slot disposed on the housing, a drive gear operatively connected to the manually engageable portion, and a driven gear operatively connected to the reel.

12. The retractable rule assembly of claim 11, wherein the manually engageable portion includes a rest configuration and an operational configuration, wherein, when the manually engageable portion is in its rest configuration, there is no engagement between the drive gear and the driver gear, and wherein, when the manually engageable portion is in its operational configuration, the drive gear and the driver gear are configured to engage with each other so as to rotate the reel.

13. The retractable rule assembly of claim 12, wherein, when a force is applied on the manually engageable portion, the manually engageable portion is moved along the slot from its rest configuration to its operational position, and wherein, when the manually engageable portion is in its operational configuration, the drive gear and the driver gear are configured to engage with each other to facilitate the rotation of the reel and the advancement of the blade in the direction that extends the blade outwardly from the housing.

14. The retractable rule assembly of claim 13, wherein the manually engageable portion includes a spring-biased manually engageable lever.

15. A retractable rule assembly comprising: a housing; a reel rotatably mounted in the housing; an elongated blade wound on the reel and constructed and arranged to extend outwardly through an opening in the housing; a coil spring configured to rotate the reel in the housing assembly in a direction to wind up the elongated blade when extending outwardly of the housing; and a spring biased blade advancer, the spring biased blade advancer comprising a pusher and a spring, the blade advancer being constructed and arranged such that when actuated, the spring pushes the pusher to engage a portion of the blade or the reel against a bias of the coil spring to advance the blade in a direction that extends the blade outwardly from the housing.

16. The rule assembly of claim 15, wherein the blade comprises a hook, and wherein the pusher, when moved by the spring, contacts the hook to extend the blade outwardly from the housing.

17. The rule assembly of claim 15, wherein a spring force of the blade advancer spring is sufficiently large to overcome the spring force of the coil spring retracting the blade into the housing.

18. The rule assembly of claim 15, wherein the blade advancer remains extended until manually forced to a retracted position.

19. The rule assembly of claim 18, wherein the blade advancer serves as a shock absorber against rapid retraction of the blade while in the extended position.

20. The rule assembly of claim 19, wherein the blade advancer serves as additional finger support while in the extended position.
